Tekendra Bhatt, Ph.D.

Augusta, GA • (469) 558-0824
bhatt_tekendra@hotmail.com • tbhatt@augusta.edu
tekendra-bhatt.github.io

Education

- **Ph.D. in Mathematics (Probability & Stochastic Processes)** 2026
  University of North Texas, Denton, TX
  Dissertation: Estimation and Statistical Inference for Hawkes Processes
  Advisor: Dr. Kai-Sheng Song.
- **M.A. in Economics** 2012
  Kumaun University, Nainital, India.
- **B.Ed. in Mathematics** 2006
  Tribhuvan University, Kathmandu, Nepal.
- **M.A. in Mathematics** 2004
  Tribhuvan University, Kathmandu, Nepal.
- **B.A. in Mathematics & Economics** 2002
  Tribhuvan University, Kathmandu, Nepal.

Academic Appointments and Teaching Experience

Assistant Professor of Mathematics August 2026 - Present
Augusta University, Augusta, GA

- Teach College Algebra (MATH 1111) and Precalculus (MATH 1113), with responsibility for instruction, assessment, grading, and student support.
- Scheduled to teach Elementary Statistics (MATH 1401) and Precalculus (MATH 1113) in Spring 2027.
- Incorporate comprehensive group projects into College Algebra and Precalculus to reinforce concepts and encourage collaborative learning.
- Offer individual meetings and maintain an open-door approach to address learning difficulties and support students' academic progress.
- Use D2L Brightspace and Pearson MyLab Math to organize course materials, assignments, and feedback.

Teaching Fellow / Teaching Assistant 2018 - 2026
University of North Texas, Denton, TX

- **Instructor of Record:** Taught Precalculus, in person and online, and Business Calculus; responsible for lectures, assessments, grading, and student support.
- **Teaching Assistant:** Led Calculus I and II recitation sections of fewer than 40 students within coordinated courses, aligning instruction with common lectures and assessments.
- **Student Support:** Provided regular office hours, individualized feedback, and early outreach to students experiencing difficulties.
- **Mathematics Tutoring:** Provided individual support in the Mathematics Learning Lab (MathLab), covering courses from Precalculus to Differential Equations.
- **Teaching Recognition:** Received the Department of Mathematics *Outstanding Teaching Fellow Award* in 2023.

Assistant Professor of Mathematics 2015 - 2018
Far Western University, Nepal

- Taught undergraduate and graduate courses, including Calculus, Differential Equations, Linear Algebra, Real Analysis, Topology, Abstract Algebra, Business Mathematics, and Statistics.
- Guided undergraduate course projects designed to reinforce mathematical concepts and deepen student understanding.

Mathematics Faculty 2010 - 2015
Brixton College and Kanchan Vidya Mandir College, Nepal

- Taught high-school mathematics for Grades 11 - 12.
- Taught Business Mathematics and Business Statistics for BBA and BBS programs.
- Taught classes of approximately 40 - 100 students, using organized lessons, practice activities, and regular assessment.

Mathematics Faculty 2005 - 2010
Bal Jagritee College, Nepal

- Taught Calculus, Linear Algebra, and Real Analysis for B.Ed. programs.

Teaching Interests

Foundational and Core Mathematics:
College Algebra, Precalculus, Calculus I - III, Business Calculus, Linear Algebra, Differential Equations, Discrete Mathematics, Introduction to Proofs.

Probability, Statistics, and Computation:
Probability, Introductory Statistics, Mathematical Statistics, Stochastic Processes, Statistical Computing with R, Introduction to Data Science, Financial Mathematics.

Advanced and Applied Mathematics:
Real Analysis, Abstract Algebra, Complex Analysis, Topology, Vector Calculus, Numerical Analysis, Mathematical Modeling.

Research Interests

- **Applied Probability:** Hawkes processes, self-exciting point processes, oscillatory excitation kernels, stability, and ergodicity.
- **Statistical Inference:** Maximum likelihood estimation, asymptotic theory, identifiability, Fisher information, and model diagnostics.
- **Moment Methods:** Moment and cumulant calculations for point processes using differential operators and related analytic methods.
- **Computation and Applications:** Simulation and estimation in R, reproducible analysis, and statistical modeling of event-time data.

Manuscripts in Preparation

- **Non-monotonic Hawkes Processes: Theory, Inference, and Applications**
  Manuscript in preparation.
- **Analytic Moment Formulas for Hawkes Processes via Differential Operators**
  Manuscript in preparation.
- **Utilizing the Hawkes Process to Apply Orthogonality in Normed Linear Space**
  Collaborative manuscript ready for submission.

Research Software

Software Package for Simulation and Inference in Hawkes Processes (R)

- Developing tools for simulation, parameter estimation, and model diagnostics; documentation in progress.

Research Experience

Ph.D. Researcher 2022 - 2026
University of North Texas, Department of Mathematics

- Developed oscillatory Hawkes models with sinusoidally modulated, exponentially decaying excitation kernels.
- Established conditions for stability, ergodicity, and identifiability of the proposed models.
- Proved consistency and asymptotic normality of maximum likelihood estimators under appropriate regularity conditions.
- Derived analytic expressions and approximations for Fisher information and assessed their accuracy through Monte Carlo studies.
- Implemented simulation, estimation, and diagnostic methods in R for reproducible analysis and model comparison.
- Applied the models to high-frequency financial event data to compare oscillatory and monotonically decaying excitation kernels.
- Developed a differential operator framework for moment calculations with exponential, gamma-type, and oscillatory Hawkes kernels.

Presentations and Proposed Student Research

When Events Breed Events: A Capstone Project Pitch 2026
Augusta University, presentation to master's students.

- Developed and presented a proposed capstone project on Hawkes process modeling and probabilistic forecasting, with potential applications to neural spike trains and cybersecurity event logs.
- Outlined a semester plan covering simulation, estimation, comparison with a Poisson baseline, forecast evaluation, and a final report and presentation.

Course Development

Simulation and Estimation of Point Processes 2026
Proposed MATH 6980 special topics course, Augusta University.

- Submitted a course proposal at the graduate director's invitation for graduate committee consideration.
- Proposed topics include homogeneous and inhomogeneous Poisson processes, basic linear Hawkes processes, event-time simulation, likelihood-based estimation, and simulation studies.
- Proposed coursework includes implementation in R or Python and a computational project.

Professional Service and Leadership

Augusta University 2026 - Present

- **Member, College Algebra and Precalculus Core Committees:** Prepare common examinations, review questions, coordinate course content, and contribute to assessment policies.
- **Member, CLASS Faculty Liaison Committee:** Collaborate on connecting classroom instruction with academic support, including early intervention, study strategies, and student persistence.
- **Member, Mathematics Contest Committee:** Serve on the committee for the university's annual mathematics contest.

Previous Service

- **Instructional Design, UNT:** Adapted department-provided Canvas course shells for synchronous and asynchronous instruction, organizing weekly modules, resources, and assessments.
- **Examination and Assessment, Far Western University:** Worked with the Office of the Controller of Examinations to prepare and review final examinations and coordinate grading policies.
- **Curriculum Development, Far Western University:** Served on the undergraduate curriculum committee and contributed content for the Topology course.
- **University Activities:** Served on committees for sports and co-curricular activities and hosted student orientation programs and internal workshops.
- **Event Leadership:** Served as Master of Ceremonies for the university convocation and other academic events.

Awards and Honors

- Outstanding Teaching Fellow Award, Department of Mathematics, UNT 2023
- Nominee, University-wide Outstanding Teaching Fellow Award, UNT 2023
- Academic Excellence Award, Department of Mathematics, UNT 2022

Research Support and Scholarships

- Summer Research Assistant Award, College of Science, UNT 2024, 2025
- Texas Public Education Grant-International (TPEG-I), UNT 2024
- Tuition Benefit Program, Graduate School, UNT 2018 - 2026
- R. Mauldin Scholarship, Department of Mathematics, UNT 2021 - 2022

Technical and Teaching Tools

- **Statistical Computing:** R, R Markdown, Mathematica.
- **Mathematical Writing:** LaTeX.
- **Teaching Technology:** D2L Brightspace, Pearson MyLab Math, Canvas, WebAssign, Respondus, Microsoft Forms, Desmos, and GeoGebra.

References

Available upon request.